Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic · 720 trials
Total training paths	1 · 720 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/NewDynamicReports/RF_Models

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	Accel
Sensor columns	72 channels
Classes	6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap

Sensor grid (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✓	✓	✓	✓
left index	✓	✓	✓	✓
left middle	✓	✓	✓	✓
left ring	✓	✓	✓	✓
left pinky	✓	✓	✓	✓
left wrist	✓	—	—	—
right palm	✓	—	—	—
right thumb	✓	✓	✓	✓
right index	✓	✓	✓	✓
right middle	✓	✓	✓	✓
right ring	✓	✓	✓	✓
right pinky	✓	✓	✓	✓
right wrist	✓	—	—	—

Enabled: 44 · Disabled: 0 (channels per enabled IMU depend on modality flags above).

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 3888
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=2880 · test=144

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M- = macro-averaged, W-* = support-weighted. Prec / Rec / F1 on hold-out test.*

Algorithm	CV acc	CV std	CV F1	CV F1 σ	Test	M-Prec	M-Rec	M-F1	W-Prec	W-Rec	W-F1
Random Forest	1.0000	0.0000	1.0000	0.0000	0.9931	0.9933	0.9931	0.9931	0.9933	0.9931	0.9931

Per-class metrics (hold-out test)

=== Random Forest ===

	precision	recall	f1-score	support
Double_ClosedHand_Flip_OpenHand	1.00	1.00	1.00	24
Double_FingerTips	1.00	1.00	1.00	24
Double_Okay_Ring_Middle_Pinky	1.00	0.96	0.98	24
Double_OpenHand_Flip_ClosedHand	1.00	1.00	1.00	24
Double_OpenHand_LPunch_R_Punch	0.96	1.00	0.98	24
Double_Snap	1.00	1.00	1.00	24
accuracy			0.99	144

macro avg	0.99	0.99	0.99	144
weighted avg	**0.99**	**0.99**	**0.99**	**144**

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.8917

Per-class external accuracy

Class	Random Forest	Support
Double_ClosedHand_Flip_OpenHand	0.900	20
Double_FingerTips	0.450	20
Double_Okay_Ring_Middle_Pinky	1.000	20
Double_OpenHand_Flip_ClosedHand	1.000	20
Double_OpenHand_LPunch_R_Punch	1.000	20
Double_Snap	1.000	20